

MSBX 5405

Structured Data Modeling and Analysis

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus, Online Section Only)	AI Engineer (85.4 / 100)	68.1 / 100

68

CURRENT AVG SCORE
across 6 job roles

98

POTENTIAL SCORE
with recommended additions

+42

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Explores both the functional and technical environment for the creation, storage and use of the most prevalent source and type of data for business analysis, ERP and related structured data. Students will learn how to access and leverage information via SQL for analysis, aggregation to visualization, create dashboards, and be source for business intelligence.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	85.4	100.0	+14.6	LOW
Machine Learning Engineer	76.5	100.0	+23.5	MEDIUM
Data Scientist	68.5	100.0	+31.5	HIGH
ML Engineer	64.4	91.6	+27.2	MEDIUM
Applied Scientist	60.7	100.0	+39.3	HIGH
MLOps Engineer	53.2	95.7	+42.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

AI Engineer

85.4 → 100.0 (+14.6 pts)

perception models learning build software model systems platform

Machine Learning Engineer

76.5 → 100.0 (+23.5 pts)

sets learning analytics tools exploratory mobile design development

Data Scientist

68.5 → 100.0 (+31.5 pts)

sets analytics mitigation social build problems

ML Engineer

64.4 → 91.6 (+27.2 pts)

discovery learning machine systems models model build problems

Applied Scientist

60.7 → 100.0 (+39.3 pts)

biology lab discovery scientific computational research problems models

MLOps Engineer

53.2 → 95.7 (+42.4 pts)

detection model text systems learning machine preprocessing image

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.